

Family Metro Map: Comparison with the Algorithm of Korst, Pronk & van Wijk (2020)

1 Did the paper give a proper algorithm?

Only partly. The paper defines two problems: a *decision problem* (does a crossing-free map exist for a given shrinking factor α ?) and an *optimization problem* (find the map that maximizes $C(M) = (c_1, c_2)$). It offers two ways to solve them:

- **Exhaustive enumeration.** Every map is fixed by one left/right choice at each bend and each T-junction, so there are $2^{n_{bp}+n_{tj}}$ maps to try. This finds the true optimum but is only practical for small trees.
- **Clockwise, a heuristic for large trees.** It is greedy and breadth-first and never revisits an earlier generation. Within a generation it handles pairs of neighbouring lines in clockwise order: it tries all 2^m T-junction combinations, then searches the bends in a sliding window.

The paper gives no polynomial-time exact algorithm and no complexity result. Clockwise comes with no guarantee: on their test set $F(4,3,2)$ it found a crossing-free map in only 57% of cases at $\alpha = 1$. The only guarantee is Lemma 4.3, which says that a small enough α always makes a crossing-free map possible. The paper even leaves open whether trees smaller than 22 persons can be impossible to draw at $\alpha = 1$.

2 How my version differs

Aspect	Paper	My implementation
Input	A binary family tree: each metro line is a couple plus their children, with partners and descendants marked	Any undirected tree. Each tree edge is one link, and “continuing a line” is implicit
Child order	Ordered by age along the line	Not ordered. Larger subtrees are placed first
Gender	Stored as data for each person	Derived from the vertex label (first character’s code)
Directions	Continue straight ($g = 0$) or T-junction ($g = \pm 2$), with a $\pm 45^\circ$ bend when genders differ	Same, plus diagonal branches ($\pm 45^\circ, \pm 135^\circ$) for vertices with more than 3 children, which the paper’s binary model never needs
Link length	$\alpha^{\text{line generation}}$	Same, plus a 34 px minimum so vertex disks never touch
Search order	Clockwise, over pairs of neighbouring lines	Breadth-first, per vertex, with a beam search (width 4) over all its children’s choices together
Crossings	A hard constraint: crossing maps are pruned as infeasible	A large penalty instead, so the algorithm always produces a drawing, even when no crossing-free one exists

Aspect	Paper	My implementation
Cost	Lexicographic (c_1 = minimum vertex distance, c_2 = average distance)	A weighted score: conflicts first, then clearance (counting vertices that sit too close to a line), room ahead for each subtree, metro-style preferences, and a small average-distance term like c_2
Overlap definition	Lines may not share any point	Stricter: also flags edges that come within 6px of each other, and edges leaving a vertex in the same direction
Fixing leftovers	Suggested only: the user flips subtrees by hand, dummy nodes could help, or a smaller global α	Automated, budgeted local search: force the branch side and bend of candidate links (and their ancestors), rescale single links, or shrink a single subtree (a local version of Lemma 4.3)

On the “Directions” row: the paper’s rule is applied fully generally, so straight versus bent depends on whether the direction suits the child’s gender. In the paper’s own cases this gives the same result as its rule.

3 Additions not in the paper

Step-by-step BFS drawing, live overlap lists with NEW markers, a summary of what the fixers changed, root selection, an interactive viewer, and import/export.

4 What I left out or weakened

These are worth knowing when comparing against the paper:

- **No exhaustive optimum.** Small trees could be solved exactly by trying all maps, but this is not implemented.
- **No family semantics.** Partners, family lines and age ordering are not modelled, so the layout is not a strict family metro map in the paper’s sense.
- **Fixers can break the paper’s spacing.** When they rescale a link or shrink a subtree, the paper’s rule of equal spacing within a generation no longer holds.
- **Different scoring.** The weighted score replaces the paper’s strict lexicographic $C(M)$, so the results are not directly comparable with the paper’s tables.

Reference: J. Korst, V. Pronk and J. J. van Wijk. A Visualization of Family Relations Inspired by the London Metro Map. *VINCI 2020*, Eindhoven, The Netherlands. <https://doi.org/10.1145/3430036.3430065>